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AUTOCAD AND SOLIDWORKS
FOR
MECHANICAL ENGINEERS

PROJECT TITLE

Automatic Solar Panel Cleaning System Using Camera Sensor

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DECLARATION

This is to certify that Mr. **VINAYAK VIJAYMAHANTESH KUBAKADDI** has successfully completed the four months training given in “**AutoCAD and Solid Works for Mechanical Engineering**” by SURE Trust .

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ABSTRACT

This project deals with the design and development of an Automatic Solar Panel Cleaning System Using Camera Sensor to improve the efficiency and performance of solar photovoltaic panels. Solar panels are commonly exposed to dust, dirt, bird droppings, and other environmental contaminants, which can reduce the amount of sunlight reaching the panel surface and consequently decrease power generation. The main purpose of this project is to develop an automated cleaning system that can detect dust or unwanted particles on the solar panel surface using a camera sensor and perform cleaning with minimum human intervention.

The proposed system consists of a mechanical cleaning mechanism, camera sensor, control unit, motor, relay module, power supply, and other supporting components. The camera sensor is used to capture images of the solar panel surface and monitor its condition. Based on the detected condition, the control system can activate the cleaning mechanism when cleaning is required. The cleaning mechanism moves across the panel surface and removes accumulated dust and dirt, helping to maintain a cleaner surface for better solar energy absorption.

The system can be designed and modeled using suitable mechanical design and CAD techniques to ensure proper movement, alignment, and operation of the cleaning mechanism. The prototype can be evaluated based on factors such as cleaning effectiveness, camera detection accuracy, movement of the cleaning mechanism, power consumption, reliability, and ease of operation. The system can further be improved by integrating advanced image processing, sensors, wireless communication, and automatic control algorithms for more accurate and efficient cleaning.

The proposed automatic solar panel cleaning system can be useful in solar power plants, residential solar installations, commercial buildings, industries, and other photovoltaic applications where regular cleaning is required. By reducing manual cleaning requirements, the system can help save time, reduce maintenance effort, and maintain the performance of solar panels. Overall, this project demonstrates the practical application of mechanical design, automation, camera-based monitoring, and control systems in renewable energy applications and provides a foundation for further development of an intelligent and fully automated solar panel cleaning system.

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CHAPTER 1

INTRODUCTION

Solar energy has gained increasing importance as a renewable energy source worldwide. With the growing awareness of environmental issues and the increasing demand for energy, solar energy offers a promising solution to address these challenges[1]. Solar energy continues to grow at an impressive pace. In fact, in 2022, the global installed solar energy capacity exceeded 1000 gigawatts (GW), representing an increase of approximately 25% compared with the previous year[2]. Solar energy currently accounts for about 4.5% of the total electricity production capacity worldwide. This growth is mainly attributed to the declining costs of solar installations and the increasing adoption of renewable energy support policies in many countries[3].

With the increasing adoption of renewable energy sources by both industries and residential consumers, photovoltaic solar panels have arisen as a pivotal resolution due to their global feasibility and availability. However, certain drawbacks associated with this technology, particularly concerning post-installation maintenance issues such as dust accumulation, have been identified. The settling of dirt elements on the solar surfaces of panels can steer to significant harm to their internal components[4]. The types of suspended particles vary based on the geographical location of the installed system and the surrounding environment. These particulates have a detrimental impact on the efficiency of these solar technologies, resulting in decreased energy yields and heightened operation and maintenance costs[5].

Diverse solutions exist in the present scenario, but their applicability depends on the quantity of panels involved. For instance, extensive solar parks, such as large-scale solar power plants, employ automated solar panel cleaning mechanisms[6]. While effective, these mechanisms tend to be operationally expensive, making them feasible primarily for sizable solar parks. Conversely, smaller systems rely on manual labour for cleaning, a risky endeavour in extreme conditions that poses potential harm to personnel and system integrity due to the methods employed (such as brushing that may damage panel surfaces)[7].

CHAPTER 2

2.1 LITERATURE REVIEW

Manju et al. [2018]: Automatic Solar Panel Cleaning System, India faces significant energy supply issues, with 60-70% of energy demand met by fuelwood and agriculture residues. Renewable energy sources like solar power are crucial to replace fossil fuels. Solar energy is a promising renewable source, but dust accumulation on solar panels reduces power generation capacity. In India, dust can decrease power output by up to 50% if panels aren't cleaned regularly. An automated cleaning system using Arduino programming has been designed to remove dust from solar panels, improving power efficiency. This solution aims to enhance solar energy utilization in India, contributing to a more sustainable energy future.

Minh Phu et al. [2024]: Design of solar panel cleaning system, based on the study, a prototype of solar panel cleaning device is calculated, design and manufacturing. The prototype can clean dust accumulated on PV modules to enhance the best power generation ability of the solar panels. The device has automatic mode and manual mode based on Wi-Fi control and app Blynk so that it can adapt to user's requirement. This device can be developed in the future for application into practice.

Najmi et al. [2023]: A Review on Solar Panel Cleaning Systems and Techniques, The accumulation of unwanted deposits on solar panels poses a significant challenge to their efficiency and energy production. These deposits, which can include dust, dirt, and other pollutants, block sunlight and reduce the panels' ability to convert solar energy into electricity. As a result, the panels' efficiency is compromised, leading to reduced energy output and increased maintenance costs. To mitigate this issue, various cleaning methods have been developed, including manual and automated techniques. The choice of cleaning method depends on several factors, including the size and location of the solar panel installation, the availability of resources such as water and labour, and the cost of the cleaning method. By understanding the impact of unwanted deposits on solar panels and evaluating different cleaning methods, solar panel users can optimize their maintenance strategies and maximize energy production. Effective cleaning and maintenance are essential to ensuring the long-term performance and efficiency of solar panel systems.

Phu et al. [2019]: Research, design and manufacture a prototype of solar panel cleaning system via Wi-Fi, this study presents the design and manufacturing of a solar panel cleaning system prototype in Vietnam, featuring 2D and 3D designs. The device can be controlled manually or automatically via Wi-Fi using an ESP8266 NODEMCU board and the Blynk app. This allows operators to remotely control the system, enhancing convenience and safety. The prototype aims to improve power generation efficiency by removing dust from solar panels, reducing labour and time, and ensuring safety compared to manual cleaning methods.

Patil et al. [2023]: Design and Analysis of Automated Solar Panel Cleaning System, this study developed an automated solar panel cleaning machine Remove dust particles that reduce panel efficiency, prevent damage to electrical components Ensure stable electricity generation. The machine uses high-quality microfiber cloth and offers Efficient operation, Superior cleaning performance, Cost-effectiveness.

Olorunfemi et al. [2022]: Solar Panels Dirt Monitoring and Cleaning for Performance Improvement, the use of solar panels as a renewable energy source has become increasingly important due to their reliability and cost-effectiveness. However, dirt accumulation on solar panels significantly reduces their energy production by blocking sunlight. To address this issue, smart systems have been developed to monitor and clean solar panels, optimizing their performance. Research has shown that these smart systems, which can detect dirt conditions and clean panels, outperform autonomous and manual technologies in terms of efficiency, accuracy, and reliability. Real-time monitoring and cleaning capabilities enable users to remotely track the condition of their solar panels and take control actions. While the research highlights the benefits of smart systems, it notes that further empirical analysis is needed to fully understand their potential and limitations. Overall, smart systems offer a promising solution to improving solar panel performance and energy output.

Zatsarinnaya et al. [2020]: The global energy industry is undergoing a significant transformation due to the volatility of the hydrocarbon market and growing environmental concerns. Renewable energy sources, particularly solar energy, are becoming increasingly prominent due to their availability and ease of use. However, the efficiency of photovoltaic panels (PVP) is compromised by dirt accumulation on their surface, reducing energy generation. To address this issue, a new panel cleaning system is being developed, utilizing a linear actuator and controlled by an Arduino UNO R3 microcontroller. This system aims to

minimize human intervention, ensuring timely and efficient cleaning of PVP, and ultimately enhancing energy production.

Nkinyam et al. [2025]: Design and implementation of a waterless solar panel cleaning system, this study presents a novel, waterless automatic cleaning system for small solar panels, addressing efficiency losses caused by dust accumulation. Key features include Arduino microcontroller with real-time clock for optimized cleaning schedules., Two-stage mechanism: ejector blower and flexible brush for effective dust removal., Strategic cleaning intervals for consistent maintenance without manual intervention. Test results show a significant 26.23% increase in average power output after cleaning. This system is particularly beneficial in arid regions where water conservation is crucial, promoting sustainable energy practices and enhancing operational efficiency for solar plant operators.

Balamurugan et al. [2025]: Advancements in solar panel maintenance, this study presents an innovative IoT-integrated automatic dust cleaning device for solar panels, designed to enhance their performance and longevity. The device employs a combination of dry and wet cleaning techniques, utilizing brushes to remove loose debris and a vinegar-based solution with mild detergent to tackle stubborn stains. By integrating IoT technology, the system enables proactive maintenance and remote monitoring capabilities, allowing users to access data from anywhere in the world through a cloud server. Experimental results demonstrate significant improvements in energy output and operational efficiency, with an average increase of 12.37% after cleaning. This novel approach offers a cost-effective solution for solar farm maintenance, reducing the need for frequent manual interventions and promoting sustainability in solar energy systems.

Vaghani et al. [2019]: Automated Solar Panel Cleaning System using IoT, this project introduces an innovative automatic solar panel cleaning system designed to enhance performance and efficiency for solar panel users. By utilizing the latest technology, the system provides transparency in cleaning operations, ensuring integrity and consistency. Notably, it boosts energy output by approximately 32% compared to dusty panels. The system can be controlled remotely via a global application, significantly reducing the need for manual labour in cleaning solar panels. This automatic solution is cost-effective, scalable, and offers a convenient way to maintain solar panels, ultimately improving overall performance and user experience.

Kar et al. [2020]: Design and development of waterless solar panel cleaning system, India receives abundant sunlight, with around 300 clear days annually, resulting in a solar energy incidence of approximately 5000 trillion kWh. However, solar panel efficiency suffers when they're dirty. Traditional water-based manual cleaning methods are costly, environmentally damaging, and pose electrical hazards. Our design introduces an automated solar panel cleaning robot that uses nylon-bristled brushes, gentle on panels, and self-navigating. The robot efficiently cleans solar panels without applying pressure, ensuring no damage to photovoltaic cells. This eco-friendly solution can be retrofitted to existing or customized solar panels, enhancing their performance and longevity.

Bhandari et al. [2024]: Performance Analysis of Semi-Automatic Solar Panel Cleaning System, this study highlights the importance of regular cleaning for solar panels to improve their efficiency and reliability. A semi-automatic dust-cleaning mechanism was designed and implemented, resulting in a 7% increase in efficiency. The findings emphasize the significance of maintaining clean solar panels to optimize energy production and reduce costs. By improving cleaning processes and lowering maintenance costs, solar panel owners can enhance the profitability and sustainability of their systems, promoting wider adoption of solar energy.

Ciufudean et al. [2024]: Automatic System for Solar Panel Cleaning, This article presents the development of a prototype autonomous robot designed to automatically clean solar panels, enhancing energy production efficiency. The system's primary goal is to provide a financially affordable solution for individual consumers, as existing automatic cleaning systems are often too expensive. By automating the cleaning process, the robot aims to address the limitations of current manual cleaning methods, which can be inconvenient and pose risks for service providers.

Zulfahmi et al. [2021]: Automatic Cleaning System Design to Increase PV Panel Output Cleaning of Solar Panel using Automation Technique, In tropical regions like India, solar panels are prone to dust accumulation, which can block sunlight and reduce power output by up to 50%. To address this issue, an automatic cleaning mechanism has been designed to sense and clean dust from solar panels. Utilizing a PIC 16F 1527 microcontroller, the system controls a stepper motor-driven gantry with mist nozzles that spray pressurized air and water

to effectively clean the panels. The mechanism adapts to dust levels, using air or water as needed, ensuring optimal energy production from solar PV modules.

Hamid et al. [2024]: Design of Automatic Cleaning System on Solar Panel Using IoT-based Wiper, this research developed an automatic solar panel cleaning device to improve energy efficiency by removing dust and dirt. The device uses a single-channel relay module, dust sensors, RTC module, and servo motor to clean the panel. Integrated with IoT technology, it enables real-time monitoring of the panel's condition via Android devices. The results show a significant voltage difference of 9.57 volts between clean and dirty conditions, highlighting the importance of regular cleaning. This innovation aims to reduce maintenance costs and enhance energy usage efficiency for the community.

Mobin [2014]: Design and Development of Solar Panel Cleaning System, this project focuses on improving solar power plant efficiency by developing an autonomous robotic cleaning system for solar panels. The system uses pneumatic suction cups to move on the panel surface and employs dry cleaning methods, such as rotating cylindrical brushes and vacuum cleaning, to remove accumulated dust. By reducing human involvement in the cleaning process, the system aims to enhance safety and efficiency in harsh environments. This innovation is particularly relevant for areas with limited water availability, where solar plants are often located.

Khadka et al. [2020]: Current Practices of Solar Photovoltaic Panel Cleaning System and Future Prospects of Machine Learning Implementation, This paper explores factors affecting Solar Photovoltaic (SPV) performance, including temperature, humidity, soiling, and weather patterns. Soiling, in particular, impacts power plant performance and lifespan, influencing the levelized cost of energy (LCoE). The authors discuss existing cleaning techniques and highlight the need for data-driven decision-making models to optimize cleaning interventions. They propose a schematic model incorporating machine learning and data processing techniques to improve cleaning efficiency. By integrating data science and machine learning, this approach could significantly advance renewable energy management.

María et al. [2023]: Automatic and portable cleaning photovoltaic solar panels mechanism, this research developed an automatic and portable cleaning mechanism for photovoltaic panels to mitigate the impact of dust accumulation on efficiency. The study found that dust

reduced load efficiency by an average of 11.05% and fill factor (FF) by up to 0.04 units. A clean panel achieved a maximum FF of 0.29, compared to 0.25 for a dirty panel. The cleaning mechanism is designed to be programmable, portable, and adjustable to different panel sizes, aiming to minimize power loss and maintain optimal performance.

Sivan et al. [2017]: Automatic Self-Cleaning Solar Panel, this research developed an automatic and portable cleaning mechanism for photovoltaic panels to mitigate the impact of dust accumulation on efficiency. The study found that dust reduced load efficiency by an average of 11.05% and fill factor (FF) by up to 0.04 units. A clean panel achieved a maximum FF of 0.29, compared to 0.25 for a dirty panel. The cleaning mechanism is designed to be programmable, portable, and adjustable to different panel sizes, aiming to minimize power loss and maintain optimal performance.

Prasad et al. [2020]: Illustration of Automatic Panel Cleaning System, this paper presents an innovative approach to maintaining solar panel efficiency by utilizing Artificial Intelligence (AI) and Computer Vision to detect and remove soiling, such as dust, bird droppings, and snow. A drone equipped with a camera and end effector is employed to monitor and clean the panels. Machine Vision techniques process images to assess the type of soiling, enabling the drone to take corrective action. The integration of AI and Computer Vision also provides valuable insights into factors influencing soiling, allowing for predictive maintenance and improved efficiency. This technology offers a promising solution to mitigate the impact of soiling on solar panel performance.

Balamurugan et al. [2025]: Advancements in Solar panel Maintenance, this study developed an IoT-integrated automatic dust cleaning device for solar panels, combining dry and wet cleaning techniques to enhance performance and longevity. The device effectively mitigates dust accumulation's negative impact, improving energy output and operational efficiency by an average of 12.37%. Experimental results showed that dust exposure reduced solar module efficiency by 7.32% after a day, 10.78% after a week, and 14.52% after a month. The IoT integration enables proactive maintenance and remote monitoring, offering a cost-effective solution for solar farm maintenance and reducing manual interventions. This innovative design contributes to the sustainability of solar energy systems, providing remote data access via a cloud server.

2.2 Summary of Literature study:

After studying of all research papers we finally conclude that, They all are uses different methods and mechanisms to cleaning the solar panels to improve the efficiency. Such as,

- In first paper they uses Arduino programming has been designed to remove dust from solar panels, improving power efficiency.
- In second paper they developed the device has automatic mode and manual mode based on Wi-Fi control and app Blynk so that it can adapt to user's requirement.
- In next paper the machine uses high-quality microfiber cloth and offers Efficient operation, Superior cleaning performance.
- In next research shows that smart systems, which can detect dirt conditions and clean panels, outperform autonomous and manual technologies in terms of efficiency, accuracy, and reliability.
- In next they utilized linear actuator and controlled by an Arduino UNO R3 microcontroller.
- In next they uses Two-stage mechanism such as ejector blower and flexible brush for effective dust removal.
- In next the system can be controlled remotely via a global application, significantly reducing the need for manual labour in cleaning solar panels.
- They design to introduces an automated solar panel cleaning robot that uses nylon-bristled brushes, gentle on panels, and self-navigating
- In next they developed semi-automatic dust-cleaning mechanism and implemented, resulting in a 7% increase in efficiency.

So after, considering all points we also developed Automatic solar panel cleaning mechanism based on camera sensor to clean the solar panel to improve the out put efficiency.

CHAPTER 3

3.1 PROBLEM STATEMENT:

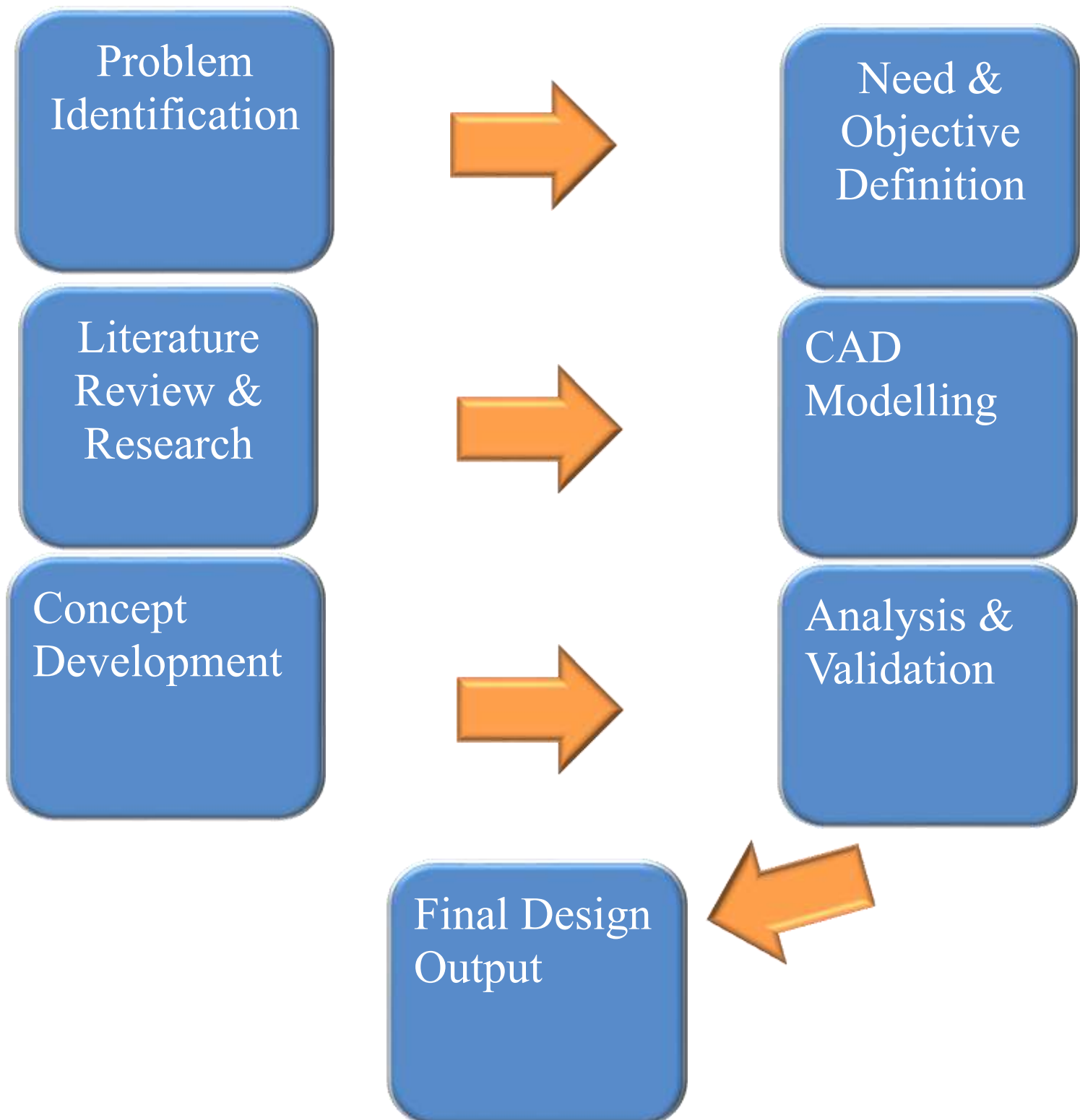
- The presence of airborne particulates settling on solar panels can significantly impact the functioning of the embedded photovoltaic cells.
- These particles obstruct the required amount of sunlight, leading to insufficient energy production.
- Additionally, the hindrance caused by these particles can result in severe damage to the silicon wafers inside.
- For instance, if dust accumulates and blocks a small area of a cell, it can cause the essential element within the cell to burn, resulting in the appearance of black spots on the panel's surface.
- Solar panels are widely used to generate clean and renewable electricity. However, dust, dirt, bird droppings, and other particles accumulate on the surface of solar panels over time. This reduces the amount of sunlight reaching the photovoltaic cells and consequently decreases the efficiency and power output of the solar panel.
- Manual cleaning requires significant time, labour, and water, especially in large solar installations. Therefore, there is a need for an efficient, economical, and automated solar panel cleaning system that can regularly remove dust and dirt from the panel surface with minimum human intervention. The proposed system aims to improve solar panel performance, reduce maintenance effort, and maintain maximum power generation through an effective cleaning mechanism.

3.2 OBJECTIVE OF THE PROPOSED PROJECT:

The objectives of this work were

- To clean the solar panel effectively.
- To make the system automated using Arduino.
- To avoid the manual work.
- To avoid dust associated problems on solar panels.
- To design and develop an automated solar panel cleaning system to remove dust, dirt, and other particles from the panel surface.
- To improve the efficiency of solar panels by maintaining a clean surface for maximum absorption of sunlight.
- To reduce manual labour and maintenance time required for cleaning solar panels.
- To develop a cost-effective cleaning mechanism suitable for domestic as well as large-scale solar installations.
- To minimize water consumption during the cleaning process.
- To ensure safe and effective cleaning without causing damage to the solar panel surface.
- To improve the overall power generation and performance of the solar energy system.
- To create a reliable system with minimum human intervention for regular solar panel maintenance.

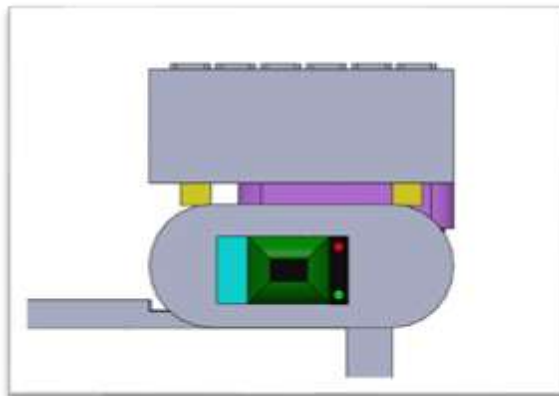
4.0 METHODOLOGY



5.0 FUNCTIONS OF COMPONENTS

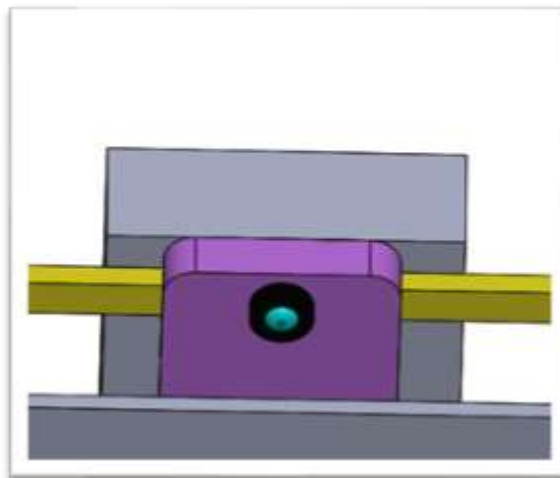
RELAY MODEL

- Turn electrical devices ON/OFF
- Control high-power devices using a low-power signal
- Provide electrical isolation between control and power circuits
- Work automatically with sensors and programmed systems
- Control devices such as motors, pumps, lights, and other electrical equipment



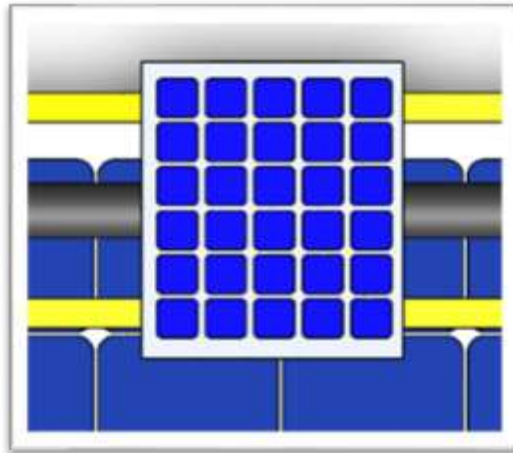
CAMERA SENSOR

- Capture photos
- Stream live video over Wi-Fi
- Detect/monitor objects (with programming)
- Send images and data through Wi-Fi
- Work with sensors and automation systems



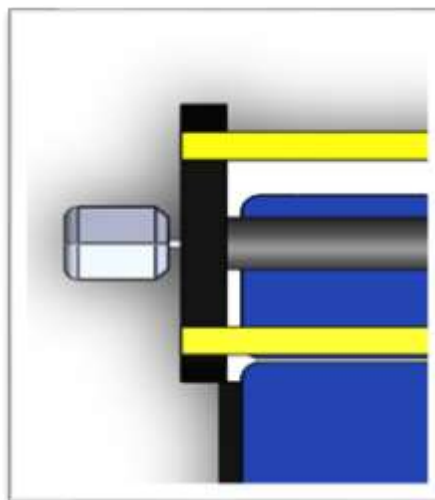
SOLAR MODEL FOR POWER SUPPLY

- Convert sunlight into electrical energy
- Supply electrical power to the system
- Charge the battery for energy storage
- Provide power to electronic components and motors
- Enable the system to operate using renewable solar energy

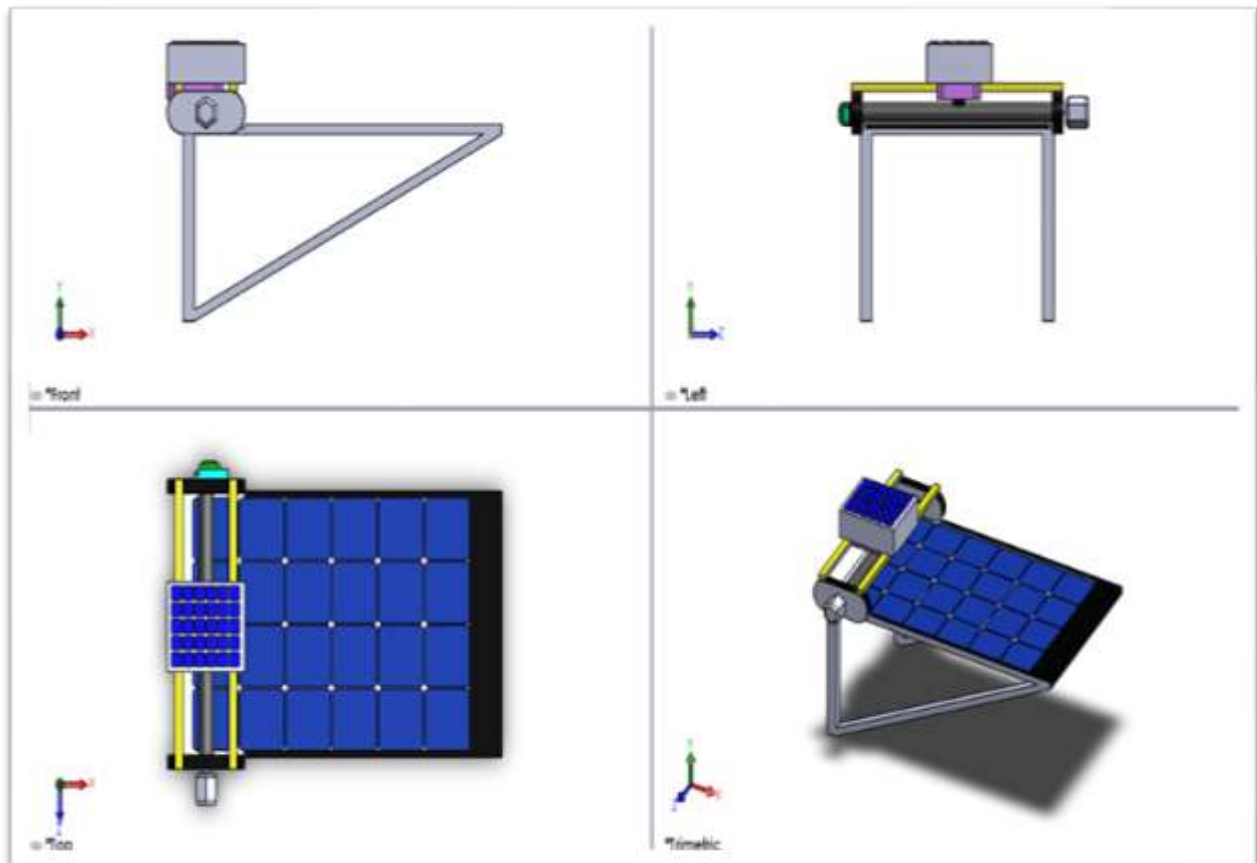


MOTOR

- Convert electrical energy into mechanical motion
- Provide rotational or linear movement for the cleaning mechanism
- Drive the brush or cleaning mechanism across the solar panel
- Work automatically based on the control system signals
- Enable efficient movement and operation of the solar panel cleaning system



6.0 ALL VIEWS OF THE MODEL



CHAPTER 7

Future Work

The future work of the solar panel cleaning system can focus on improving its automation, efficiency, and reliability. Dust and environmental sensors can be integrated to automatically detect dirt accumulation and initiate the cleaning process when required. IoT-based monitoring can allow users to control and monitor the system remotely through a mobile application. The system can also be powered using solar energy generated by the panel itself, making it more energy-efficient and sustainable. Future developments may include waterless cleaning mechanisms, weather-based control, AI-based cleaning schedules, and improved robotic movement for different sizes and orientations of solar panels. The system can also be further developed for large-scale solar power plants to clean multiple panels automatically while reducing maintenance cost and improving overall power generation efficiency.

To transform the physical model into a fully functional automated system, the following future enhancements can be implemented:

- 1. Electronic Integration** Add microcontroller-based control (ESP32/Arduino). Interfacing drivers for motors, pump, and sensors. Circuit protection and stable power distribution design.
- 2. Programming & Automation** Develop firmware that automates cleaning cycles. Include manual override and safety shutdown functions. Add error detection (motor jam, low water level, etc.).
- 3. IoT Features (Optional but powerful)** Real-time monitoring through Blynk or similar apps. Remote start/stop of cleaning cycles. Data logging for panel performance and cleaning frequency.
- 4. Mechanical Optimization** Improve brush or wiper design for better cleaning without scratching. Optimize water usage and Weatherproofing for long-term outdoor operation.

CHAPTER 8

CONCLUSION

The prototype of the automatic solar panel cleaning system has successfully demonstrated the core mechanical functionality required for efficient dust removal. With the physical model complete, the system now provides a strong foundation for integrating electronics and automation. The design ensures minimal manual intervention and aims to maintain optimal panel efficiency by preventing dust accumulation. Once the control circuitry, sensors, and programming logic are added, the prototype will evolve into a fully automated solution capable of operating with precision and reliability. This marks a significant step toward improving solar power generation, especially in dusty environments where energy loss is high.

The solar panel cleaning system is designed to provide an effective and efficient solution for removing dust, dirt, and other particles accumulated on the surface of solar panels. Regular cleaning helps maintain maximum sunlight absorption and improves the overall power generation efficiency of the solar panel. The proposed system reduces manual effort, saves time, and provides a safer and more convenient cleaning method. By using an automated mechanism, the system can perform cleaning operations efficiently and consistently. Overall, the solar panel cleaning system is a practical, cost-effective, and sustainable solution that can help improve the performance and lifespan of solar power systems.

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